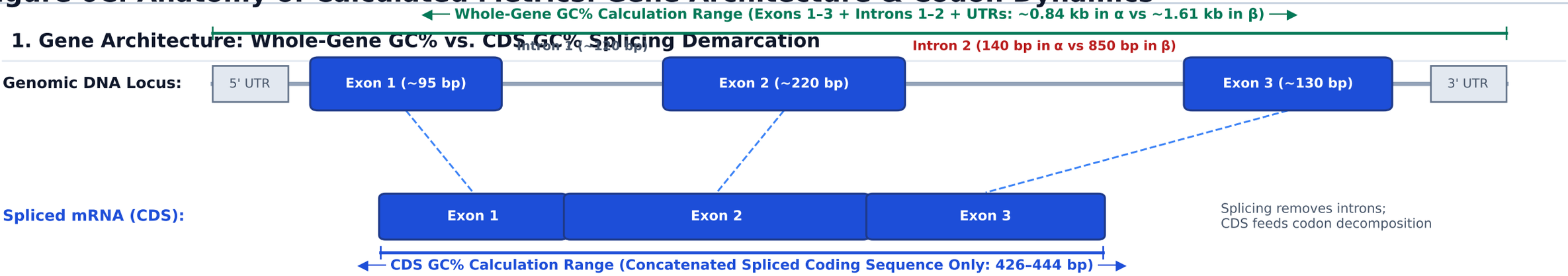
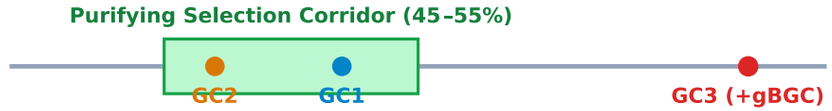
Figure 0C: Anatomy of Calculated Metrics: Gene Architecture & Codon Dynamics

1. Gene Architecture: Whole-Gene GC% vs. CDS GC% Splicing Demarcation



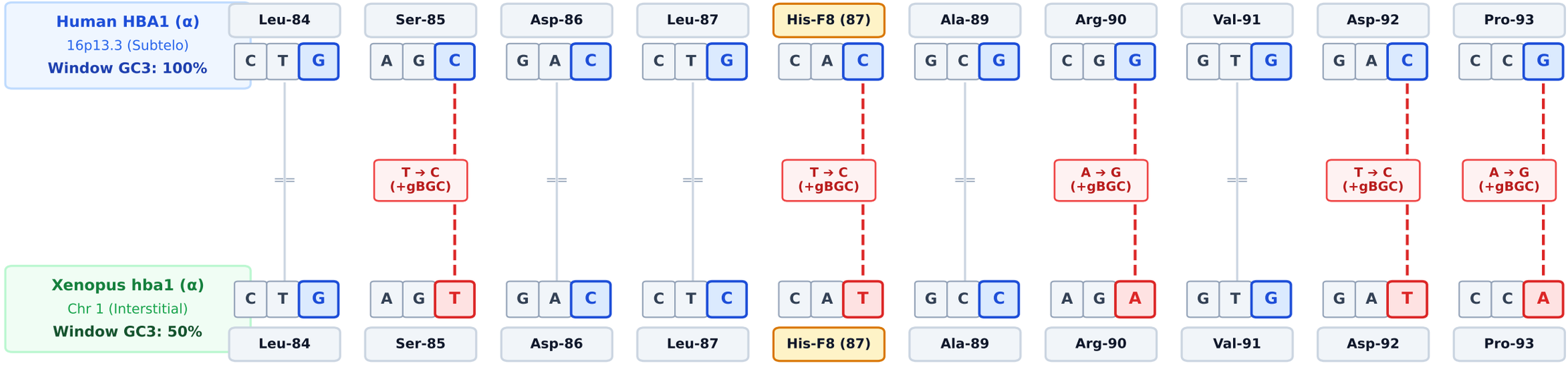
Codon Position Decomposition & Selection Constraints:

- **GC1: 1st position (~50–65% GC) — Constrained by amino acid polarity**
- **GC2: 2nd position (45–55% GC) — Strictly locked in Purifying Selection Corridor across all 68 vertebrates**
- **GC3: 3rd position (35–98% GC) — Synonymous site! Neutral barometer of GC-biased gene conversion (gBGC)**



2. Conserved Globin Heme Pocket: Proximal Histidine (His-F8) Motif & Synonymous GC3 Divergence

Comparison across 350+ Myr: 100% Identical Amino Acids, Yet 100% of Divergent 3rd Positions Shifted to G/C in Human



3. Bernardi's Compaction Law & Telomeric Distance Driver

Bernardi's Compaction Law: High-GC isochores suppress intron expansion; low-GC regions accumulate expanded introns.

Human HBA1 (α): 843 bp (Intron 2 = 140 bp) [Compact High-GC] vs. Human HBB (β): 1,608 bp (Intron 2 = 850 bp) [Expanded Low-GC]

Distance-to-Telomere (dtelo / Lchr) Drives gBGC:

• Meiotic crossovers are physically biased toward chromosome ends, accelerating GC-biased gene conversion (gBGC).

• Subtelomeric loci (dtelo < 1 Mb) undergo runaway GC3 elevation; internal/centromeric loci remain buffered in AT-rich baselines.

`Relative Telomeric Distance: dtelo_rel = min(Pos, Chr_Len - Pos) / Chr_Len    (0.000 Subtelomeric  →  0.500 Metacentric)`